



NCCER Module 22107



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Vertical-Mast Sit-Down Counterbalance Forklifts





Section One

Forklift Safety Guidelines

Objective

1. Identify and describe how operators can safely drive a forklift.
 - a. Describe OSHA requirements for operating a forklift.
 - b. Describe the primary causes of forklift fatalities and how they can be avoided.
 - c. Describe the role equipment inspections play in forklift safety.
 - d. Identify and describe safe forklift refueling practices.
 - e. Review and identify hand and verbal signals for forklift operations.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

1. Demonstrate the proper hand signals and voice signals used in forklift operation.



1.0.0 Forklift Safety Guidelines

- Safe forklift operation is your responsibility.
- Operators must develop safe working habits and recognize hazardous conditions to protect themselves and others from injury or death.
- Always be alert for unsafe conditions that may jeopardize your safety or the safety of your coworkers.
- Before operating your forklift, be sure to familiarize yourself with the operator's manual, the controls on your forklift, and the forklift's primary safety features.
- Ensure that you are properly trained and certified.



1.1.0 Operator Training and Certification

- Only trained and qualified personnel are allowed to operate forklifts on the job.
- The operator must receive written and practical performance testing and possess certification of the training.
- Individual companies may have their own guidelines that specify how they comply with these OSHA guidelines.



1.2.0 Operator Safety (1 of 2)

- Most forklift accidents happen as a result of being struck by a forklift or **tip-overs**.
- When a worker is struck by a forklift, it is often because the driver and worker aren't aware of each other's positions.
- Tip-overs occur when the forklift's center of gravity shifts to the point that the forklift becomes unstable and turns over.



1.2.0 Operator Safety (2 of 2)

- Always wear proper PPE.
- Operate the forklift according to the **rated capacity** and manufacturer's limits.
- Keep **forks** and **attachments** lowered when traveling.
- Never operate a forklift while smoking, using a cell phone, or under the influence of intoxicants.
- Only operate the machine from the operator's cab.
- Always stay alert to your surroundings and the presence of others.



1.2.1 Co-worker Safety

- Make sure that workers are clear of the path of the forklift at all times.
- When your ability to see or hear is impaired, use a **spotter** and a radio.
- Be careful at blind spots, crossings, and other locations with heavy traffic or where pedestrians may step into your path.
- Always keep your focus in the direction of travel.



1.3.0 Equipment Inspections

- Forklift operators should regularly inspect their machines.
- Look and listen to make sure the equipment is functioning normally.
- When an equipment problem is discovered, fix it immediately.
- NEVER operate a broken forklift.



1.4.0 Refueling

- Internal combustion (IC) powered forklifts may run on LPG, gasoline, a combination of both, or on diesel.
- Electric-powered forklifts use large batteries that drive a strong motor.
- Both types require the operator to refuel (or recharge) regularly.



1.4.1 IC Forklifts (1 of 3)

Procedure for refueling gas- or diesel-powered forklifts:

1. Park the forklift in a designated refueling area.
2. Using the **F-N-R lever**, place the transmission in neutral.
3. Lower the mast/forks to the ground.
4. Set the **parking brake**.
5. Turn off the engine.
6. Open the cap to the fuel reservoir.
7. Fill the tank slowly, ensuring that it is not overfilled.
8. Close the cap when the tank is full.



1.4.1 IC Forklifts (2 of 3)

- LPG burns cleaner than gasoline or diesel.
- LPG tanks should never be replaced in low-lying areas or in confined spaces near stairwells, maintenance pits, or egress points.
- Always wear gloves to protect from frostbite.
- Ensure that the forklift's fuel line is empty before attempting to disconnect it from the tank.



1.4.1 IC Forklifts (3 of 3)

Procedure for refueling LPG powered forklifts:

1. Park the forklift in a designated refueling area.
2. Using the F-N-R lever, place the transmission in neutral.
3. Lower the mast/forks to the ground.
4. Set the parking brake.
5. With the forklift running, turn off the LPG tank valve.
6. Let the forklift run until it stops.
7. Attempt to restart it.
8. When it will not restart, disconnect the fuel line and remove the tank.
9. Install a full LPG tank – ensure it is oriented properly – and secure it.
10. Connect the fuel line and open it slowly.
11. Check for leaks using a soapy solution.



1.4.2 Electric Forklifts (1 of 2)

Procedure to charge a forklift battery:

1. Park the forklift in a designated charging area.
2. Open the battery compartment lid and ensure the area is well ventilated.
3. Ensure the forklift master (or operating) **switch** is turned off.
4. Disconnect the forklift power cable from the battery.
5. Check the battery cables for damage.
6. After confirming the charger is off, connect the battery charger cable to the battery.



1.4.2 Electric Forklifts (2 of 2)

Procedure to charge a forklift battery (continued):

7. Turn on the charger and and charge the battery until it is at full charge.
8. When the battery is at full charge, turn off the charger.
9. Disconnect the battery charger cable from the battery.
- 10.Reconnect the forklift power cable to the battery's connection.



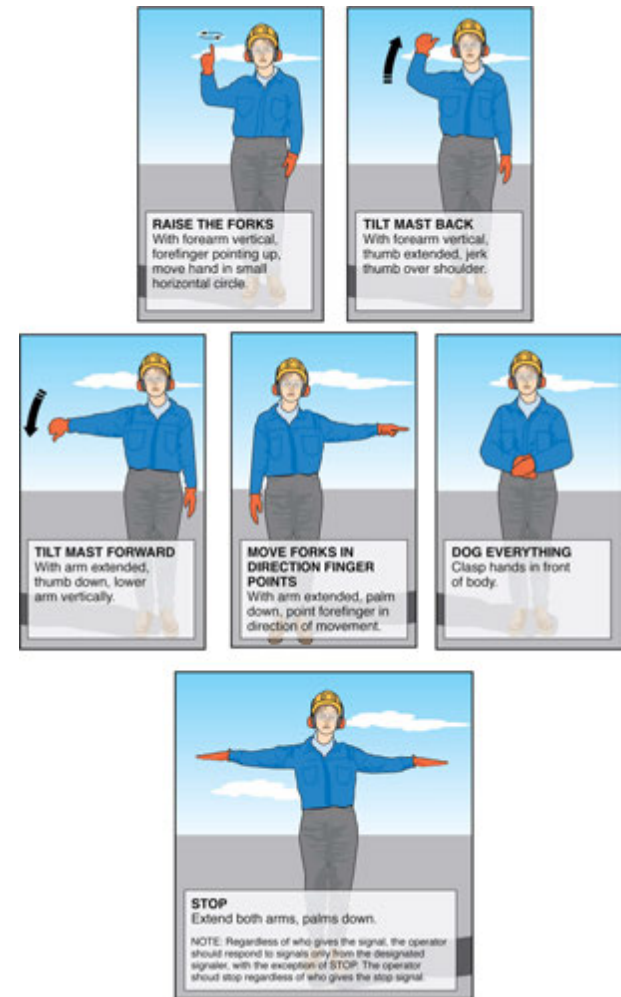
1.4.3 Spill Containment and Cleanup

- Designated refueling areas built to contain any spills.
- Check for hydraulic fluid leaks regularly.
- Clean and repair leaks as quickly as possible.
- Make sure that an oil spill kit is easily accessible.



1.5.0 Hand and Voice Signals Used with Forklifts

- Forklift operators should ask a co-worker to serve as a spotter or signal person whenever sight is restricted.
- The signal person can use electronic communication devices, voice commands, or **hand signals**.
- **Voice signals** should be agreed upon by the forklift operator and signal person prior to moving a load or maneuvering the lift.



Wrap Up – Trade Terms (1 of 3)

Attachments: Accessories used on conventional forklifts that enable them to perform jobs which could not be accomplished with forks alone.

F-N-R Lever: A lever typically located on the left side of the steering column. The operator uses this lever to place the torque-converter-driven transmission in forward, neutral, or reverse.

Forks: Flat tines connected to the forklift's carriage, upon which loads are lifted.

Hand signals: Used by a worker (signal person) to tell a forklift operator how to safely move a load or maneuver a lift.



Wrap Up – Trade Terms (2 of 3)

Parking brake: Handle used to set or release the brake. It should always be set when the forklift is unattended. A hydrostatic drive forklift usually has an electric switch that activates the brake by electronically shutting off the drive controls.

Rated capacity: The heaviest load that a forklift can carry without tipping.

Spotter: A worker who has a clear view of the forklift, load, and operator and can assist an operator whose view or hearing may be impaired.



Wrap Up – Trade Terms (3 of 3)

Switch: A switch either makes or breaks a connection. Forklift switches are mechanical or electrical and typically disconnect critical functions so that the machine cannot be operated. They are also used to turn accessories like lights on or off.

Tip-overs: Events that occur when the forklift's center of gravity shifts outside the stability triangle and the forklift becomes so unstable that it turns over.

Voice signals: Voice commands that have been agreed upon by the forklift operator and signal person prior to moving a load or maneuvering a lift.



1.0.0 Section Review Question 1

1. Due to liability concerns, insurance guidelines, and OSHA requirements, forklift operators are required to be _____.
- a. experienced
 - b. polite
 - c. trained
 - d. insured



1.0.0 Section Review Question 1 Answer

1. Due to liability concerns, insurance guidelines, and OSHA requirements, forklift operators are required to be _____.
- a. experienced
 - b. polite
 - c. trained**
 - d. insured



1.0.0 Section Review Question 2

2. Before operating a forklift, the operator should be familiar with the forklift's _____.
- a. operator manual
 - b. mast type
 - c. forklift class
 - d. fuel type



1.0.0 Section Review Question 2 Answer

2. Before operating a forklift, the operator should be familiar with the forklift's _____.

a. operator manual

b. mast type

c. forklift class

d. fuel type



1.0.0 Section Review Question 3

3. What items should be kept by the forklift operator as proof of appropriate training.
- a. certifications or course completions
 - b. emails and letters of recognition
 - c. copies of training materials
 - d. key dates when training occurred



1.0.0 Section Review Question 3 Answer

3. What items should be kept by the forklift operator as proof of appropriate training.
- a. certifications or course completions**
 - b. emails and letters of recognition
 - c. copies of training materials
 - d. key dates when training occurred



1.0.0 Section Review Question 4

4. What is a key trait that safe operators always possess when driving a forklift?
- a. calm
 - b. even-tempered
 - c. alert
 - d. patient



1.0.0 Section Review Question 4 Answer

4. What is a key trait that safe operators always possess when driving a forklift?
- a. calm
 - b. even-tempered
 - c. alert**
 - d. patient



1.0.0 Section Review Question 5

5. True or False: Regular inspections play an important role in forklift safety.
- a. True
 - b. False



1.0.0 Section Review Question 5 Answer

5. True or False: Regular inspections play an important role in forklift safety.

a. True

b. False



1.0.0 Section Review Question 6

6. Before refueling a forklift, the operator should _____.

- a. keep the engine running and fill the tank to the full mark
- b. remain on the forklift and set the parking brake
- c. turn the engine off and avoid smoking
- d. put on gloves and wipe up any spilled fuel



1.0.0 Section Review Question 6 Answer

6. Before refueling a forklift, the operator should _____.
a. keep the engine running and fill the tank to the full mark
b. remain on the forklift and set the parking brake
c. turn the engine off and avoid smoking
d. put on gloves and wipe up any spilled fuel



1.0.0 Section Review Question 7

7. True or False: Hand signals are nice for a forklift operator to know, but not required.
- a. True
 - b. False



1.0.0 Section Review Question 7 Answer

7. True or False: Hand signals are nice for a forklift operator to know, but not required.

a. True

b. False



1.0.0 Section Review Question 8

8. True or False: There are some situations where OSHA regulations allow passengers to ride on a forklift or be lifted on the forks.
- a. True
 - b. False



1.0.0 Section Review Question 8 Answer

8. True or False: There are some situations where OSHA regulations allow passengers to ride on a forklift or be lifted on the forks.
- a. True
 - b. False**



Next...

Section 2.0.0
Forklift Fundamentals



*Read Sections 2.0.0 through 2.2.4 and
complete the Section Review.*